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Abstract:	Differential abundance or expression analyses are routinely performed on metagenomic, (meta)transcriptomic, and amplicon sequencing data. In such datasets, analysts usually have no information regarding the true scale (i.e. size) of the microbial community or sample under study, with inter-sample differences in sequencing depth instead being driven by technical variation rather than biological factors. Recent work has demonstrated that normalisations used in all analysis tools make incorrect assumptions about the biological scale of the system in question, leading to unacceptably high false discovery rates in the output. To mitigate this, analysts can acknowledge and account for uncertainty of the overall system scale during normalisation by building scale models of the data- a feature that has been integrated into the ALDEx2 R package. Here, we use provide reproducible examples that demonstrate how to incorporate scale models into differential expression analyses of RNA-seq data using bulk transcriptome and metatranscriptomic datasets, as well as the consequences of not doing so. We also show how to use the output of ALDEx2 to create high-level, exploratory visualisations of their data through principal component analysis.

Incorporating scale uncertainty into differential expression analyses using ALDEx2

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ABSTRACT:

Basic Protocol 1: Using a simple scale model for differential expression analysis to avoid 'dual cutoff' P value / significance thresholds

Basic Protocol 2: Implementing a full, informed scale model to correct scale-related data asymmetry in differential expression analyses

Basic Protocol 3: Visualising ALDEx2 outputs using compositional approaches: principal component analysis

Differential abundance or expression analyses are routinely performed on metagenomic, (meta)transcriptomic, and amplicon sequencing data. In such datasets, analysts usually have no information regarding the true scale (i.e. size) of the microbial community or sample under study, with inter-sample differences in sequencing depth instead being driven by technical variation rather than biological factors. Recent work has demonstrated that normalisations used in all analysis tools make incorrect assumptions about the biological scale of the system in question, leading to unacceptably high false discovery rates in the output. To mitigate this, analysts can acknowledge and account for uncertainty of the overall system scale during normalisation by building scale models of the data- a feature that has been integrated into the ALDEx2 R package. Here, we use provide reproducible examples that demonstrate how to incorporate scale models into differential expression analyses of RNA-seq data using bulk transcriptome and metatranscriptomic datasets, as well as the consequences of not doing so. We also show how to use the output of ALDEx2 to create high-level, exploratory visualisations of their data through principal component analysis.

KEYWORDS:

Differential expression; Differential abundance; RNA-seq; Metagenomics; ALDEx2

INTRODUCTION:

High-throughput sequencing (HTS) is a popular tool for investigating differences in composition and/or gene expression between microbial communities or cell populations. Commonly used approaches include bulk, single-cell or meta transcriptome profiling by RNA-seq for gene expression analysis; or amplicon sequencing of universal 'barcoding' genes; or shotgun metagenomic sequencing (Manzoni et al., 2016). At each sampling step of the laboratory work, from the initial DNA/RNA extraction to library construction/multiplexing and loading onto a flow cell, critical information regarding the scale of the community is lost (e.g. total number of nucleic acid fragments, total gene expression levels or overall microbial load). The final output from a sequencing experiment is usually a table of counts of genes or other features across multiple samples. These counts represent a small subset of the final DNA library loaded onto the sequencer, which is itself a subset of the individually prepared sequencing libraries, which is itself a subset of the original DNA extract, and so on, all the way back to the sample taken from the original community. Importantly, the instrument also imposes an upper limit on the number of molecules that can be sequenced (Gloor et al., 2017). Therefore, information contained in the sequencing output only informs on the relative abundances of the genes/features. Scale information can be obtained by complementary approaches alongside sequencing, including: estimating total bacterial loads via qPCR or droplet digital (dd)PCR of the 16S rRNA gene (Barlow et al., 2020); 'spiking in' a known number of intact cells or molecules which can be used to back-normalise relative abundance values (Rao et al., 2021); quantifying the total number of bacterial cells using flow cytometry (Y. Lin et al., 2019); and estimating the number of viable cells through microbial cultivation and CFU/mL determination (Rolling et al., 2021). These approaches are not routinely employed in most microbiome studies, however.

Analysis of HTS data is inherently variable, due to the lack of information about the absolute size, or "scale". Notably, the false-discovery rate (FDR) is often poorly controlled by many popular tools and inference of differentially expressed genes can be subjective because it is routinely based upon non-standard cutoffs (e.g. fold-change) (Ebrahimpoor & Goeman, 2021; Gloor et al., 2025; Li et al., 2022). Such approaches have been shown to inadvertently inflate the rate of false discoveries (Bourgon et al., 2010; Ebrahimpoor & Goeman, 2021). Likewise, many sequencing datasets reflect biological situations where there is an innate difference in the overall mean difference between the conditions of interest, or there may be a prevalence difference that manifests as a sparse data table where sets of genes are present in samples from one condition but not the other (Nixon et al., 2024). Multiple tools perform poorly on such datasets (Li et al., 2022; Gloor et al., 2025). Often in HTS analyses, scale itself is a major confounder of differential expression/abundance and recognition of this in the field of transcriptomics drove the development of normalisation methods (Bullard, Purdom, Hansen, & Dudoit, 2010). Examples of scale issues that complicate HTS analyses pervade the literature: total and mRNA levels are increased approximately 2-fold and 3.5-fold, respectively, in B lymphocytes overexpressing the oncogene, cMyc, compared to non-transformed cells (Nie et al., 2012). However, follow-up investigations by the same group determined that this could drastically impact transcriptome analysis unless a spike-in normalisation was performed (Lovén et al., 2012). Similarly, total bacterial loads within the vaginal microbiome differ between states of health and bacterial vaginosis by approximately 10 to 100-fold (Zozaya-Hinchliffe et al., 2010) and failing to account for this leads to many false-positive and false-negative findings during differential expression analysis (Dos Santos et al., 2024; Gloor et al., 2025). Finally, differing growth rates and RNA contents have been observed between wild type and mutant strains of bacteria, fungi and eukaryotic cell lines which, as above, hinder one's ability to infer truly differential genes from scale-related artefactual findings (J. Lin & Amir, 2018; Scott et al., 2010; Yoshikawa et al., 2011).

When conceptualising the scale problem in HTS data, Nixon and colleagues demonstrated that all tools used for differential expression/abundance analysis make varying assumptions regarding scale during normalisation and that these assumptions are demonstrably wrong (Nixon et al., 2024). This is one major reason that different normalisations often return vastly different results when applied to the same input data. The original ALDEx2 model is guilty of the same: an inherent and incorrect assumption is made relating the geometric mean of the compositional data to the system scale (Fernandes et al., 2014; Nixon et al., 2024). To address this shortcoming, Nixon *et al*. used a series of simulations and algebraic proofs to show that modifying this assumption by introducing uncertainty around the scale (i.e. by modelling) could greatly improve both FDR control and reproducibility of the results. ALDEx2 builds a Bayesian posterior model of the HTS count data using Monte-Carlo sampling and estimates the underlying technical variation (i.e. what one would see if the same experiment was repeated many times) (Fernandes et al., 2014). These imputed data are subject to a centred log-ratio (CLR) transformation before testing the null hypothesis that mean log-ratio differences between groups are not different, with the results averaged across the Monte Carlo instances. Modelling of scale uncertainty was incorporated into the log-ratio transformation step, and this explicitly accounts for the fact that any individual normalisation is wrong, but that the uncertainty provides an estimate of many reasonable normalisations. The exact details of scale modelling in ALDEx2 are discussed elsewhere (Gloor et al., 2025; McGovern et al., 2023; Nixon et al., 2024).

Here, we describe the steps used to perform differential expression analyses on several example datasets using the scale-aware ALDEx2 package, which can be applied to any other HTS dataset. We use a highly replicated yeast transcriptome SNF2 knockout dataset to show how 'simple' scale models can be used to eliminate the need for dual P-value and fold-change cutoffs which are commonly used to limit lists of differentially expressed genes down to a 'manageable' number of genes likely to be truly differential. Further, we use a vaginal metatranscriptome dataset which represents a 'worst-case scenario' (where there are vast differences in scale and gene content between conditions) to show how a more complex 'full' scale model can be constructed to overcome this issue. Finally, we demonstrate how to determine the effect of increasing amounts of scale uncertainty on a given dataset and provide insight on how to select an appropriate value.

BASIC PROTOCOL 1

Basic protocol title:

Using a simple scale model for differential expression analysis to avoid 'dual cutoff' *P* value / significance thresholds

Introductory paragraph:

In a highly replicated yeast transcriptome dataset generated by Gierliński and colleagues, a majority of genes (65-80 %) are identified as differential by commonly used, scale-naïve tools such as DESeq2, edgeR and ALDEx2 (using the original algorithm as in (Fernandes et al., 2014)). This dataset comprises 48 lab (growth technical) replicates of wild-type *Saccharomyces cerevisiae* vs. a Δ*snf2* mutant. In a benchmarking study of differential expression tools using the same data, the authors recommended employing a fold-change threshold to the list of significantly differentially expressed genes to control for potential false discoveries (i.e. genes with a low/marginal fold-change between conditions with *P* <0.05). There are no universally agreed upon values for either threshold and therefore different researchers are free to choose their own *P* value or log fold-change cutoffs, often using volcano plots, which is known to inflate the FDR further (Ebrahimpoor & Goeman, 2021). In this protocol, we provide an example of how adding a small amount of scale uncertainty into the ALDEx2 model can achieve a similar result of controlling for false positive findings without the need for arbitrary thresholds.

Materials:

A feature table containing the number of reads per feature (i.e. gene/taxon) across every sample in your dataset (here we use the data from Gierliński *et al*. ('data/yeast_counts.txt', from this study's GitHub repository). Samples should be in columns; features should be in rows.

A metadata table containing group membership information (here we use the data from Gierliński *et al*. ('data/yeast_metadata.txt', from this study's GitHub repository)

Access to a computer with R and RStudio installed and >8GB RAM, ALDEx2 and its dependencies must be installed from Bioconductor (pre-computed ALDEx2 outputs are provided for replication of this protocol as an alternative for those who do not meet this requirement).

For this protocol, R (v4.5.1) and RStudio (2024.12.1, build 563) were used, along with the following R packages: BiocManager (v1.30.26), ALDEx2 (v1.41), dplyr (v1.1.4), ggplot2 (v3.5.2), and patchwork (v1.3.1).

Protocol steps with *step annotations:*

Environment setup

1. Install the ALDEx2 package from Bioconductor. Note this needs only be performed once

```
if (!require("BiocManager", quietly = TRUE))

    install.packages("BiocManager")

BiocManager::install("ALDEx2")
```

2. Load the ALDEx2 package, and load in the count data and metadata tables directly from GitHub

```
library(ALDEx2)

url.gene <-
"https://github.com/scottdossantos/currprotSDS/raw/refs/heads/main/data/yeast_counts.txt"
```

```
yst.gene <- read.table(url.gene, sep = "\t", header = T, quote = "", row.names = 1)

url.meta <-
"https://github.com/scottdossantos/currprotSDS/raw/refs/heads/main/data/yeast_metadata.txt"

yst.meta <- read.table(url.meta, sep = "\t", header = T, quote = "", row.names = 1)
```

Differential expression analysis with ALDEx2

3. Set the seed value (required for exact replication of results on GitHub) and run the centred log-ratio transformation with half a standard deviation of scale uncertainty, specifying group membership, the type of denominator to use in the geometric mean calculation (all features), and the number of Monte-Carlo instances to generate.

```
set.seed(2025)

yst.5.clr <- aldex.clr(reads = yst.gene, conds = yst.meta$group, mc.samples = 128, denom =
"all", gamma = 0.5, verbose = TRUE)
```

For further information on the gamma parameter, see the Critical Parameters heading of the Commentary section

4. For each feature, calculate the median dispersion of log-ratio abundances within groups and the median difference in log-ratio abundances between groups (this is performed for all 128 Monte-Carlo instances, and a median of medians is returned).

```
yst.5.clr.e <- aldex.effect(clr = yst.5.clr, verbose = TRUE, include.sample.summary = TRUE)
```

When `include.sample.summary` *is set to* `TRUE`, the *output of aldex.effect will return the median log-ratio abundance values of each feature per-sample (i.e. averaged across all 128 Monte-Carlo instances), which can be used to plot a compositional biplot following principal component analysis.*

5. Conduct parametric and non-parametric tests to identify features for which the mean difference in the log-ratio abundance between groups is significantly different, adjusting for multiple testing using the Benjamini-Hochberg correction.

```
yst.5.clr.t <- aldex.ttest(clr = yst.5.clr, verbose = TRUE)
```

For applicable study designs, it is possible to specify paired t-tests by adding the call, `paired.test = TRUE`

Visualisation of differential expression analysis output

6. Visualise the output of ALDEx2 by creating volcano (difference in log-ratio abundance between groups vs. $-\log_{10}$ *P* values) and effect plots (difference in log-ratio abundances between groups vs. dispersion of log-ratio) for this analysis. ALDEx2 has a built-in function for creating basic versions of these plots (not shown), but we also show the same data plotted using the ggplot2 R package in Figure 1 (see GitHub for ggplot2 plotting code).

```
yst.5.clr.all <- rbind(yst.5.clr.e, yst.5.clr.t)

aldex.plot(yst.5.clr.all, type = "volcano")

aldex.plot(yst.5.clr.all, type = "MW")
```

BASIC PROTOCOL 2:

Basic protocol title:

Implementing a full, informed scale model to correct scale-related data asymmetry in differential expression analyses

Introductory paragraph:

In a study of vaginal metatranscriptomes from healthy vs. bacterial vaginosis (BV) patients, Dos Santos *et al*. showed how this data represents a 'worst-case' scenario of scale-related problems in omics data analysis (Dos Santos et al., 2024). This dataset consists of 42 samples from 2 separate studies: 20 from healthy or BV patients from Canada (Macklaim & Gloor, 2018), and 22 from German BV patients pre- and post- successful treatment for BV with metronidazole (Deng et al., 2018). Analysts must contend with i) differing total bacterial loads of approximately two orders of magnitude (Zozaya-Hinchliffe et al., 2010); and ii) vastly different genomic content in the metatranscriptomes between healthy and BV patients (France et al., 2022). These compositional and scale differences make it challenging to determine differentially expressed genes within this dataset. It is possible to reduce the impact of these scale-related issues by analysing this data one organism at a time; however, this precludes system-wide analyses which underpin the rationale of using meta-omics approaches. When these data are analysed as an entire system, many housekeeping functions were present within the list of differentially expressed genes which is unintuitive given our understanding of molecular biology and the routine use of such genes as invariant reference sequences or proteins in qPCR or Western blotting experiments (Joshi et al., 2022). In this protocol, we show how analysts can implement an informed scale model of their data with ALDEx2, using housekeeping functions to inform on the scale difference between groups.

Materials:

A feature table containing the number of reads per feature (i.e. gene/taxon) across every sample in your dataset (here we use the data from Dos Santos *et al*. ('data/mts_counts.txt', from this study's GitHub repository). Samples should be in columns; features should be in rows.

A metadata table containing group membership information (here we use the data from Dos Santos *et al*. ('data/ mts_metadata.txt', from this study's GitHub repository).

A way of identifying the functions of each gene or transcript contained in the feature table (here we use the KEGG orthology terms and their corresponding pathways provided by the authors of the non-redundant VIRGO database of vaginal microorganisms, 'data/mts_virgo_KOs.txt' and 'data/mts_KOlookup.txt', from this study's GitHub repository).

Access to a computer with R and RStudio installed and >8GB RAM (pre-computed ALDEx2 outputs are provided for replication of this protocol as an alternative for those who do not meet this requirement).

For this protocol, R (v4.5.1) and RStudio (2024.12.1, build 563) were used, along with the following R packages: BiocManager (v1.30.26), ALDEx2 (v1.41), CoDaSeq (v0.99.7), dplyr (v1.1.4), ggplot2 (v3.5.2), and patchwork (v1.3.1).

Protocol steps with *step annotations:*

Environment setup

1. Install the ALDEx2 package from Bioconductor. Note this needs only be performed one time

```
if (!require("BiocManager", quietly = TRUE))
    install.packages("BiocManager")
BiocManager::install("ALDEx2")
```

2. Load the required libraries, gene count table, metadata and functional lookup tables directly from GitHub.

```
library(ALDEx2)

url.gene <-
"https://github.com/scottdossantos/currprotSDS/raw/refs/heads/main/data/mts_counts.txt"

mts.gene <- read.table(url.gene, sep = "\t", header = T, quote = "", row.names = 1)

url.meta <-
"https://github.com/scottdossantos/currprotSDS/raw/refs/heads/main/data/mts_metadata.txt"

mts.meta <- read.table(url.meta, sep = "\t", header = T, quote = "", row.names = 1)
```

```
url.func <-
"https://github.com/scottdossantos/currprotSDS/raw/refs/heads/main/data/mts_KOlookup.txt"

mts.func <- read.table(url.func, sep = "\t", header = T, quote = "", row.names = 1)

url.virg <-
"https://github.com/scottdossantos/currprotSDS/raw/refs/heads/main/data/mts_virgo_KOs.txt"

mts.virg <- read.table(url.virg, sep = "\t", header = F, quote = "", row.names = 1)
```

Aggregate gene count data by function

3. Match the gene IDs to the corresponding functions in the gene vs. KEGG orthology term lookup table

```
ko.virgo <- mts.virg[[1]][match(rownames(mts.gene), rownames(mts.virg))]
```

4. For each KEGG orthology term in the dataset, sum all read counts for every gene in the dataset assigned to a given orthology term (clean up this dataframe after aggregating).

```
mts.ko <- aggregate(mts.gene, by = list(ko.virgo), FUN = sum)

rownames(mts.ko) <- mts.ko$Group.1

mts.ko$Group.1 <- NULL
```

This example uses KEGG orthology terms; however, you may use whichever functional classification system you prefer- as long as you have a way of identifying functions in your system which you know (or have reason to believe) are invariant between groups.

Determine background scale values per group

5. Set the seed value (required for exact replication of results on GitHub), then run a scale-naïve ALDEx2 centred log-ratio transformation on the data using an incredibly small (i.e. virtually no) scale uncertainty.

```
set.seed(2025)

scale.0.clr <- aldex.clr(reads = mts.ko, conds = mts.meta$group, mc.samples = 128, denom =
"all", gamma = 1e-3, verbose = TRUE)
```

For further information on the gamma parameter, see the Critical Parameters heading of the Commentary section

6. Isolate the scale estimates per Monte-Carlo instance from the log-ratio transformation output and calculate the mean background scale values.

```
bg.scale.0 <- scale.0.clr@scaleSamps

bg.scale.h.0 <- mean(rowMeans(bg.scale.0)[which(mts.meta$group == "Healthy")])

bg.scale.bv.0 <-mean(rowMeans(bg.scale.0)[which(mts.meta$group == "BV")])

abs(bg.scale.h.0 – bg.scale.bv.0)

2^abs(bg.scale.h.0 – bg.scale.bv.0)
```

The '@scaleSamps' element of the log-ratio transformation output contains all scale estimates used for each sample across all Monte-Carlo instances: the mean of the rows for all samples belonging to a given condition corresponds to the background scale values for each sample. For this example, there is a log_2 difference in scale between groups of 3.005442, equating to roughly an 8-fold overall difference in scale between samples from healthy patients vs. BV patients.

7. Subset the functional feature table (i.e. table of counts per KEGG orthology term across all samples) to extract only those related to housekeeping functions. The explicit assumption being made here is that housekeeping functions should be roughly at the same scale.

```
hk.func <- which(mts.func$pathway %in% c("Ribosome","Glycolysis /
Gluconeogenesis","Aminoacyl-tRNA biosynthesis"))

mts.ko.hk <- mts.ko[hk.func,]
```

8. Run a centred log-ratio transformation on these housekeeping using the same parameters as the full dataset.

```
set.seed(2025)

hk.clr <- aldex.clr(reads = mts.ko.hk, conds = mts.meta$group, mc.samples = 128, denom =
"all", gamma = 1e-3, verbose = TRUE)
```

9. Calculate the mean background scale estimate per group as a ratio

```
bg.scale.hk <- hk.clr@scaleSamps

bg.scale.h.hk <- mean(rowMeans(bg.scale.hk)[which(mts.meta$group == "Healthy")])

bg.scale.bv.hk <- mean(rowMeans(bg.scale.hk)[which(mts.meta$group == "BV")])

bg.scale.h.hk / bg.scale.bv.hk
```

For this example, the ratio is 1 : 1.141893, so we can infer that the difference is approximately 14 % between groups. We will use this ratio in our informed scale model to account for the scale difference between healthy and BV groups

Differential expression analysis with an informed scale model

10. Set the seed value once more and use an inbuilt ALDEx2 function to generate a matrix of scale uncertainty values to be used in the model. These values will be drawn from a log-normal distribution with a standard deviation of 0.5 and will differ by (on average) 14 % between groups.

```
set.seed(2025)

scale.f <- aldex.makeScaleMatrix(gamma = 0.5, mu = c(1, 1.14), conditions = mts.meta$group,
mc.samples = 128)
```

11. Run the centred log-ratio transformation on the full, functional dataset, specifying a custom matrix of scale values to be used during the modelling of system scale.

```
scale.f.clr <- aldex.clr(reads = mts.ko, conds = mts.meta$group, mc.samples = 128, denom =
"all", gamma = scale.f, verbose = TRUE)
```

12. For each feature, calculate the median dispersion of log-ratio abundances within groups and the median difference in log-ratio abundances between groups (this is performed for all 128 Monte-Carlo instances, and a median of medians is returned).

```
scale.f.clr.e <- aldex.effect(clr = scale.f.clr, verbose = TRUE, include.sample.summary =
TRUE)
```

13. Conduct parametric and non-parametric tests to identify features for which the mean difference in the log-ratio abundance between groups is significantly different, adjusting for multiple testing using the Benjamini-Hochberg correction.

```
scale.f.clr.t <- aldex.ttest(clr = scale.f.clr, verbose = TRUE)
```

Visualise differential expression output

14. Visualise the output of ALDEx2 by creating an effect (difference in log-ratio abundances between groups vs. dispersion of log-ratio) for this analysis. ALDEx2 has a built-in function for creating basic versions of this plot (not

shown), but we also show the same data plotted using the ggplot2 R package in Figure 2 (see GitHub for ggplot2 plotting code).

```
scale.f.clr.all <- cbind(scale.f.clr.e, scale.f.clr.t)

aldex.plot(scale.f.clr.all, type = "MW")
```

BASIC PROTOCOL 3

Basic protocol title:

Visualising ALDEx2 outputs using compositional approaches: principal component analysis

Introductory paragraph:

There are a multitude of methods for visualising and summarising HTS data. Principal co-ordinates analysis (PCoA), metric/non-metric multidimensional scaling (MDS/nMDS), uniform manifold approximation and projection (UMAP), and t-distributed stochastic neighbour embedding (t-SNE) are frequently used methods in the study of microbiomes. Our preferred approach is to conduct principal component analysis (PCA) using the centred log-ratio transformed abundances output by ALDEx2 and the `prcomp()` function built into R. This equates to calculating the Aitchison distance (i.e. the Euclidean distance between samples following CLR transformation): a truly linear distance metric which is sub-compositionally coherent and robust to subsetting or aggregation of the underlying data (Aitchison, 1986; Gloor et al., 2017). In this protocol, we outline how to perform an exploratory analysis of HTS data using the output of ALDEx2 to produce a compositional biplot.

Materials:

Combined ALDEx2 outputs of your data from `aldex.clr()`, `aldex.effect()`, and `aldex.ttest()`. Here, we use pre-computed data from Dos Santos *et al*., 'data/mts_scaleFull.Rda', from this study's GitHub repository (this is equivalent to the combined data frame produced in step 14 of Basic Protocol 2).

Access to a computer with R and RStudio installed and >8GB RAM, ALDEx2 and its dependencies must be installed from Bioconductor (pre-computed ALDEx2 outputs are provided for replication of this protocol as an alternative for those who do not meet this requirement).

For this protocol, R (v4.5.1) and RStudio (2024.12.1, build 563) were used, along with the following R packages: BiocManager (v1.30.26), ALDEx2 (v1.41), CoDaSeq (v0.99.7), dplyr (v1.1.4), ggnewscale (v0.5.2), and ggplot2 (v3.5.2).

Protocol steps with *step annotations:*

Environment setup

1. Follow steps 1-6 of Basic Protocol 1 or steps 1-14 of Basic Protocol 2 to produce a combined ALDEx2 output from the `aldex.clr()`, `aldex.effect()`, and `aldex.ttest()` functions. If replicating this example, load in the pre-computed, combined ALDEx2 output for the KO-aggregated vaginal metatranscriptomes in Dos Santos *et al*.

```
url.scaleFull <- "https://github.com/scottdossantos/currprotSDS/raw/refs/heads/main/data/mts_scaleFull.Rda"

load(url(url.scaleFull))
```

2. Extract the centred log-ratio abundance values for all features and samples from the ALDEx2 output (these columns all start with 'rab.sample.'), then remove the 'rab.sample.' from the column names.

```
scale.f.lr <- scale.f.clr.all[, grep("rab.sample.", colnames(scale.f.clr.all))]
```

```
colnames(scale.f.lr) <- gsub("rab\\.sample\\.", "", colnames(scale.f.lr))
```

3. Perform the principal component analysis using the prcomp() function.

```
pca.f <- prcomp(t(scale.f.lr))
```

Input must be transposed as the function expects samples in rows and features in columns. Note that performing PCA on CLR-transformed data is equivalent to building an ordination based on Aitchison distances (Euclidean distances between samples following CLR-transformation).

4. To plot a very basic compositional biplot, use the built in plot() function.

```
biplot(pca.f, cex = c(0.55, 0.25), col = c("blue", rgb(0,0,0,0.1)))
```

5. For more customisation, the CoDaSeq package has a more advanced plotting function specifically for visualising HTS data. As an example, we will plot a compositional biplot which highlights the KEGG pathways that several differentially expressed KO terms belong to, with all other KO terms plotted as transparent points. Start by making a list that contains the indices of the samples in the grouping vector.

```
ind.grp <- list(Healthy = which(mts.meta$group == "Healthy"), BV = which(mts.meta$group ==
"BV"))
```

To colour samples or features, the `codaSeq.PCAplot()` function requires a list which contains the groups of data you want to highlight (e.g. treatment groups, disease vs. control for sample labels; taxa or pathway membership for features) and the column/row indices of each sample or feature in the input data.

6. Make a second list that contains the indices of the KO terms to highlight in the KO to pathway lookup table, grouped by pathway.

```
pathways <- c("Aminoacyl-tRNA biosynthesis","Bacterial chemotaxis","Butanoate metabolism",
              "CAMP resistance","Exopolysaccharide biosynthesis","Flagellar assembly",
              "Porphyrin metabolism","Ribosome","Starch and sucrose metabolism",
              "Two-component system")

ind.load <- list()

for(i in pathways){ind.load[[i]] <- which(mts.func$pathway == i)}
```

7. Add all remaining KO terms to a list element called 'Other'. These will be displayed as transparent grey points.

```
ind.load[["Other"]] <- setdiff(1:nrow(scale.f.lr), unlist(ind.load))
```

8. Make two vectors of R-recognised colours, one for groups and another for pathways.

```
cols.group <- c("dodgerblue2", "orange2")

cols.load <- c("skyblue1","red3","gold2","chocolate4","greenyellow","purple3",
               "olivedrab","black","maroon","cyan3", rgb(0,0,0,0.075))
```

9. Plot the compositional biplot with `codaSeq.PCAplot()`, adding density plots for the highlighted pathways, a title, and a legend for the pathways. Samples will be coloured by group and KO terms will be coloured by pathway.

```
codaSeq.PCAplot(pca.f, plot.groups = T, plot.loadings = T, plot.density = "groups",
                PC = c(1,2), grp = ind.grp, grp.col = cols.group, grp.cex = 0.75,
                load.grp = ind.load, load.col = cols.load, load.sym = 19,
                load.cex = 0.6, plot.legend = "loadings", leg.position = "bottomright",
                leg.columns = 2, leg.cex = 0.625,
                title = "Vaginal metatranscriptome: PCA plot (\u03b3 = 0.5, \u03bc = 14 %)")
```

10. Alternatively, if complete control over the plots is desired, extract the scaled coordinates for PC1 and PC2 from the PCA object for both the samples and KO terms. These will be used as input data for ggplot2.

```
load.f.samp <- data.frame(pca.f$x)[,1:2]
```

```
load.f.feat <- codaSeq.PCAvalues(pca.f)[,1:2]
```

The data in the PCA object giving the sample scores (`pca.f$x`) corresponds to the sample coordinates on the PCA plot; however the loadings (`pca.f$rotation`) do not correspond to the coordinates of the features. This CoDaSeq function scales the loadings to the PCA scores, allowing you to plot the same biplot however you like.

11. Add pathways to the data frame containing the feature coordinates.

```
for(i in 1:length(ind.load)){load.f.feat[ind.load[[i]], "path"] <- names(ind.load)[i]}
```

12. Add a column for setting the outline of the data point (either transparent grey or solid black) based on whether the KO should be highlighted or not.

```
load.f.feat$col <- case_when(load.f.feat$path == "Other" ~ rgb(0,0,0,0.05), .default = "black")
```

13. Convert the pathway column to a factor and re-order the levels to ensure that KO terms belonging to the 'Other' pathway are plotted last (as they are transparent).

```
lvs <- levels(factor(load.f.feat$path))

load.f.feat$path <- factor(load.f.feat$path, levels = c(lvs[1:6], lvs[8:11], lvs[7]))
```

14. Add group information and a plot title to the data frame containing the sample coordinates.

```
for(i in 1:length(ind.grp)){load.f.samp[ind.grp[[i]], "group"] <- names(ind.grp)[i]}

load.f.samp$title <- "Vaginal metatranscriptome: PCA plot (\u03b3 = 0.5, \u03bc = 14 %)"
```

15. Make a vector to 1) ensure the order of the group colours; and 2) change the legend symbol for the samples (text) from an 'a' to a more appropriate symbol (in this case, 'h' and 'v').

```
grp_labels<- c(Healthy = "h", BV = "v")
```

16. Plot the compositional biplot using ggplot2.

```
ggplot(data = load.f.samp, aes(x = PC1, y = PC2))+
  geom_hline(yintercept = 0, colour = "grey50", linewidth = 0.5, linetype = 2)+
  geom_vline(xintercept = 0, colour = "grey50", linewidth = 0.5, linetype = 2)+
  geom_text(aes(colour = group), label = rownames(load.f.samp))+
  scale_colour_manual(name = "Group", values = cols.group, breaks = names(grp_labels),
                      guide = guide_legend(override.aes = list(label = grp_labels)))+
  new_scale_colour()+
  geom_point(data = load.f.feat, aes(fill = path, colour = col),
             stroke = 0.25, shape = 21, size = 2)+
  scale_fill_manual(name = "Pathway", values = cols.load,)+
  scale_colour_manual(name = "Pathway", values = c(rgb(0,0,0,0.05), "black"))+
  xlab("PC1: 49.4 % variance explained")+
  ylab("PC2: 11.0 % variance explained")+
  guides(colour = "none")+
  theme_bw()+
  facet_wrap(~title)+
  theme(strip.text = element_text(size = 9, face = "bold"),
        panel.grid.major = element_blank())
```

COMMENTARY:

Critical Parameters:

The most important parameter for these differential expression analyses is the value passed to the `gamma` argument of `aldex.clr()`, which corresponds to the standard deviation of the log-normal distribution that the scale estimates are drawn from. Therefore, `gamma` essentially represents the degree of scale uncertainty that ALDEx2 will model.

Consider the following example of volcano plots showing the ALDEx2 analysis of the yeast transcriptome dataset (Figure 1): when running a scale-naïve analysis (i.e. the model does not accept any uncertainty when estimating system scale), 4,169/5,891 genes (70.7 %) are called as significantly different between wild-type and Δ*Snf2* mutant cells (Figure 1A). Only 189 of these are considered 'truly' different when implementing a secondary $\log_2$ fold-change threshold of 1.4 in either direction (the approximate mid-point of the high and low fold-change thresholds proposed by the original authors (Gierliński et al., 2015; Schurch et al., 2016)). By implementing a simple scale model and setting `gamma` to 0.5 (i.e. the log-normal distribution that the scale estimates are drawn from has a standard deviation of 0.5), only 175 genes are called as significantly different (Figure 1B), achieving the same result as a fold-change cutoff without the need for arbitrarily deciding what that threshold should be and unintentionally increasing the FDR in the process. Figures 1C and 1D show effect plots for the corresponding analyses of the yeast dataset; when accounting for scale uncertainty, many features with a high within-group dispersion are no longer reported as differentially expressed.

In some cases, a simple scale model with a single gamma value is insufficient for proper analysis of a dataset. This issue is clearly demonstrated in Figure 2A, which shows an effect plot (log dispersion vs. log difference) of the vaginal metatranscriptome dataset in a scale-naïve ALDEx2 analysis. Positive log-difference values indicate a higher expression in samples from healthy patients while negative values indicate higher expression in samples from BV patients. Points in yellow are housekeeping functions which are largely universal and are expected to be invariant between groups (e.g. glycolysis / tRNA biosynthesis / ribosomal functions) (Joshi et al., 2022). However, the expression of many of these genes is also deemed to be significantly different (red points; n = 1,089), pointing to a problem during the normalisation. Adding a modest amount of scale uncertainty with a simple scale model does little to address this problem (Figure 2B). The dispersion of all data points has increased slightly, and a few likely false-positive findings are now no longer significantly different (n = 934; compare points just inside of the grey lines of equivalence for dispersion and difference in Figure 2A and 2B). However, the bulk of these housekeeping functions are still estimated to be significantly overexpressed in healthy metatranscriptomes. These housekeeping functions are only correctly centred around the line of no difference when an informed scale model is applied, using a scale difference between groups of 14 % (Figure 2C). This value corresponds to the average background scale estimate for each group, as determined in steps 6 – 9 in Basic Protocol 2. When the full, informed model is used, the total number of differentially expressed genes drops to 875, eliminating over 200 false discoveries relative to the initial analysis. Comparing the scale-naïve and informed scale models, one can see many false-positive and false-negative findings are present in the unscaled output when uncertainty around the system scale is not accounted for during normalisation.

The above two examples then beg the question: how does one choose an appropriate gamma value? Unfortunately, there is no simple, concrete answer. Larger gamma values will reduce the number of features called as significantly different, but one can be more confident that features identified as differential are truly so. Conversely, lower gamma values or scale-naïve analyses will return a larger number of differential features but with the risk that a proportion of these features represent false-positive findings. We have previously argued that there is no such thing as a "free lunch" when it comes to statistical analysis of -omics data (Gloor et al., 2025), and that this extends to tools beyond ALDEx2, which will always force the analyst into a trade-off, prioritizing either sensitivity or good FDR control, but never both. These observations have been replicated by others in one of the most comprehensive benchmarking studies of over 15,000 samples from 35 studies (Konnaris et al., 2025). We have also further demonstrated this in recent benchmarking work comparing ALDEx2 to several widely used differential expression analysis tools such as DESeq2, EdgeR and Limma (Dos Santos et al., 2025).

ALDEx2 contains built-in functions to allow researchers to identify features which are particularly sensitive to increasing values of `gamma` (i.e., those which were already on the margin of significance without accounting for even small amounts of scale uncertainty). The `aldex.senAnalysis()` function takes a vector of numeric values and repeats the differential expression analysis sequentially using each of the supplied values for the `gamma` argument. Likewise, the `plotGamma()` function creates a visualisation showing the effect sizes of all features in the dataset at each specified gamma value, with significance at a given gamma value indicated by colour (i.e. shows how increasing amounts of scale uncertainty affect the strength of findings in the differential expression analysis). An example output of the `aldex.senAnalysis()` function is shown in Figure 3A for the yeast transcriptome dataset: adding even a minor amount of scale (0.1) reduces the number of significantly expressed genes between wild-type and *Snf2* knockout cells by 50 %. Many of these genes were likely only significant due to marginal significance or difference values (Figure 1A). Accordingly, we recommend `gamma` values between 0.2 and 0.5 but ultimately, the analyst has a choice to make about the extent to which they wish to tolerate false discoveries.

To further inform the choice of `gamma` value to use, we can ask the question: "what is the smallest log difference between groups that will result in a significant result [P <0.05] at a given value of gamma". Figure 3B shows a linear regression of the minimum log difference required for significance vs. increasing gamma values for the yeast transcriptome dataset. As the degree of scale uncertainty increases, so too does the log difference needed to identify a feature as significantly different between conditions. Notably, the slope of the regression line is approximately 2.5, meaning that analysts can expect a given gamma value to require a log difference between their biological conditions of interest roughly two-and-a-half times as large as the `gamma` value they passed to `aldex.clr()` during normalisation. We have found that relationship between gamma and the minimum log-difference required for a significant result is relatively consistent, even when testing different types of -omics data, including bulk RNA-seq, single-cell RNA-seq, and metatranscriptome data (Dos Santos et al., 2025). Accordingly, analysts should consider how much of a difference exists between their biological conditions of interest: if there is a large and evident difference between conditions (e.g. immediately obvious when visualising beta diversity), a larger gamma value will be appropriate. In the opposite case where there is not much difference between conditions, a small value will suffice. In either case, modelling scale uncertainty with ALDEx2 will offer excellent control of the false-discovery rate at any given gamma value. We recommend at minimum that users employ a gamma value of 0.2, equating to a minimum log2 fold-change of 0.5 (i.e. a ~1.4-fold difference between groups) for any significant result. This gives an excellent control of the FDR while maintaining sensitivity and removes the need for any dual-cutoff approach.

In addition to non-parametric significance tests, ALDEx2 can also construct generalised linear models (GLMs) via the `aldex.glm()` and `glm.effect()` functions. Scale models can be used in these functions in the exact same manner as outlined in both protocols above: through the `gamma` parameter when calling `aldex.clr()`. The successor to ALDEx2, the aptly named ALDEx3, is also currently in development and is based on linear models. ALDEx3 permits modelling of scale uncertainty in the same manner as its predecessor; however, it is much faster and is no longer bound by memory constraints when analysing large datasets. For researchers looking to employ linear models in their work, ALDEx2's GLM function remains an option until ALDEx3 is formally released.

Troubleshooting Table:

Table 1: Troubleshooting guide for running ALDEx2 on any -omics dataset.

Problem	Possible cause(s)	Solution
Memory-related errors while running ALDEx2 in R	Machine used for running ALDEx2 has insufficient RAM	Lower the number of Monte-Carlo instances used to model the underlying variation in count data (controlled by the `mc.samples` argument in the `aldex.clr()` function). Alternatively, run ALDEx2 on a server with sufficient RAM.
ALDEx2 reports mismatch between number of samples and conditions vector	Count data table is in the opposite orientation (i.e. samples in rows and features in columns)	Confirm that count data has samples in columns and features in rows. Check that the count data table has been imported into R correctly and has the expected number of rows and columns (e.g. feature names are given in an actual column in the data frame rather than set to the row names attribute)
ALDEx2 reports no significant differences between groups	Gamma value is too high Incorrect grouping vector supplied	Perform a sensitivity analysis using a range of gamma values from low to high to visualise the effect of adding scale uncertainty on ALDEx2 outputs.

		Confirm that grouping vector is correct and in the same order as the samples in the count data.

Understanding Results:

The core ALDEx2 functions, `aldex.effect()` and `aldex.ttest()` return data frames which can be combined for plotting. Below, we explain what information is contained within the columns of these dataframes.

- rab.all: The median log-ratio abundance values for each feature across all samples, itself expressed as a median calculated across all Monte-Carlo instances. These are always relative to the geometric mean value of the samples
- rab.win.A: The median log-ratio abundance values for each feature across all samples in group A, itself expressed as a median calculated across all Monte-Carlo instances.
- rab.win.B: The median log-ratio abundance values for each feature across all samples in group B, itself expressed as a median calculated across all Monte-Carlo instances.
- rab.sample.X: The median log-ratio abundance value for a given feature in sample X across all Monte-Carlo instances.
- diff.btw: The log2 fold-change between conditions, calculated as in Fernandes *et al.* 2014
- diff.win: The pooled dispersion of the feature, calculated as in Fernandes *et al*. 2014. This is analogous to a pooled standard deviation.
- effect: A standardised measure of effect sizes given by dividing 'diff.btw' by 'diff.win' for a given feature, calculated as a median across all Monte-Carlo instances.
- overlap: The proportion of effect sizes that overlap zero (i.e. the percentage of Dirichlet instances in which the effect size was in the opposite direction). An overlap of 0.5 indicates that the two distributions share the same midpoint.
- we.ep: The posterior predictive P-value from Welch's t-test for each feature.
- we.eBH: The Benjamini-Hochberg-corrected posterior predictive P-value from Welch's t-test for each feature.
- wi.ep: The posterior predictive P-value from the Wilcoxon rank sum test for each feature.
- wi.eBH: The Benjamini-Hochberg-corrected posterior predictive P-value from the Wilcoxon rank sum test for each feature.

Time Considerations:

The amount of time required to run the main functions of the ALDEx2 package for differential expression analysis will vary depending on the number of samples and number of features in the read count table, as well as the number of Monte-Carlo instances specified in the call to the `aldex.clr()` function. Table 2 shows the time taken to run the three core modular functions of the ALDEx2 package for transformation, effect size calculation, and significance testing in RStudio (all functions run on a 2023 MacBook Pro with 16 GB RAM and an Apple M2 Pro processor). ALDEx2 does support multi-core processing for the `aldex.clr()` and `aldex.effect()` functions, which is implemented via the BiocParallel package. However, due to the amount of overhead required by R to set up parallel processes, this doesn't offer too much of an improvement in speed, particularly when calculating effect sizes (Table 2). Note that all simulations were performed while no applications other than RStudio were running.

CONFLICT OF INTEREST STATEMENT:

GBG is an author and current maintainer of ALDEx2.

DATA AVAILABILITY STATEMENT:

All code required for replicating these analyses and producing the figures shown in the article can be found on this study's GitHub repository at: https://github.com/scottdossantos/currprotSDS, along with the data tables containing read counts per sample and metadata.

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FIGURE LEGENDS:

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Figure 1 Using a simple scale model eliminates the need for a dual P-value / fold-change threshold: Count data from the yeast transcriptome dataset underwent log ratio transformation using the ALDEx2 package with (**A**) and without (**B**) modelling scale uncertainty (gamma = 0.5). Effect sizes and posterior predictive P values were calculated for all features, comparing wild-type and Snf2 knockout cells.

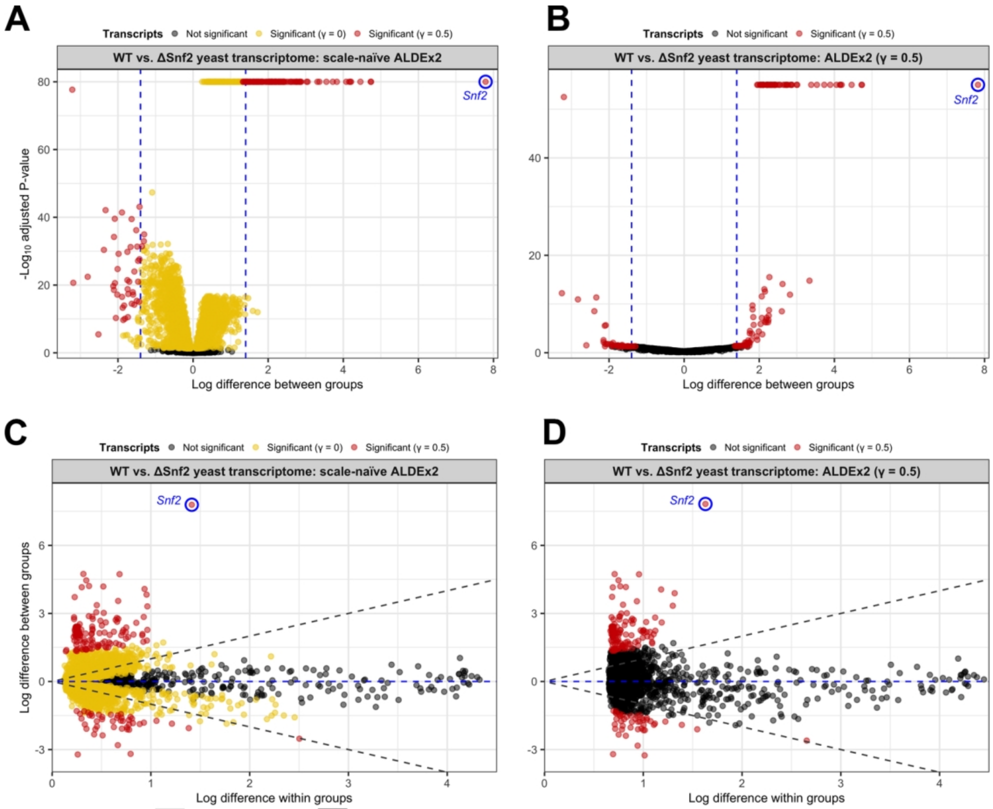
Figure 2 Implementing a full, informed scale model corrects data asymmetry in problematic datasets: Count data from the vaginal metatranscriptome dataset was aggregated by KEGG orthology term prior to log ratio transformation using three different approaches: (**A**) a scale-naïve analysis; (**B**) a simple scale model using gamma = 0.5; (**C**) a full scale model using gamma = 0.5 and a difference between conditions of 14 %. Data represented as effect plots. Black points represent non-significant functions, red points represent differentially expressed functions, yellow points represent housekeeping functions. Grey lines indicate equivalent difference within groups and difference between groups. Blue line indicates no difference between groups.

Figure 3 Sensitivity analysis informs the selection of an appropriate gamma value: Count data from the yeast transcriptome dataset was used for a sensitivity analysis in ALDEx2, using a range of gamma values from 0 to 1.0 in increments of 0.1. (**A**) Effect sizes at each gamma value are plotted for all features and coloured by significance at each gamma value. Blue numbers indicate the number of differentially expressed features at each gamma value. (**B**) The minimum absolute log difference between wild-type and Snf2 knockout cells required for a statistically significant result is shown for each gamma value. Regression line represents a slope of 2.2.

TABLES:

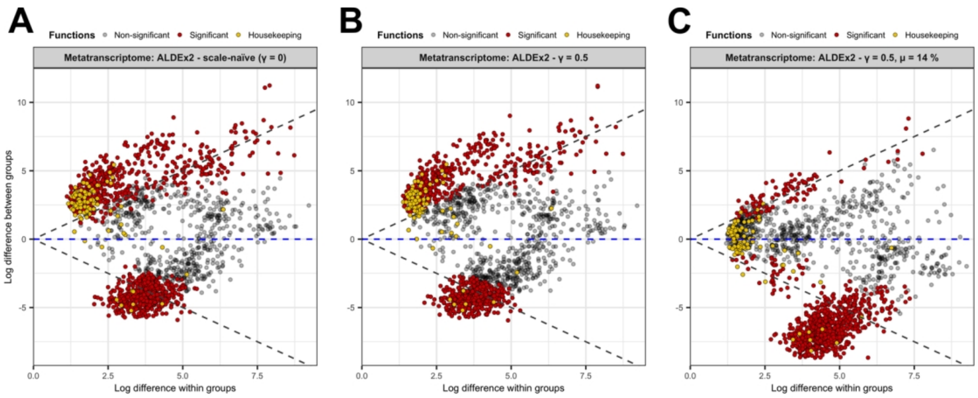
Table 2: Time taken for computation of log-ratio abundances, effect sizes and posterior predictive *P*-values when running ALDEx2 on the yeast transcriptome and vaginal metatranscriptome datasets, as well as an additional dataset: a large RNA-seq dataset from cancer cells pre- and post-exposure to the PD1 antagonist, nivolumab. Times represent a mean of 5 iterations rounded to the nearest integer, with standard deviations in parentheses. Computation was performed on a 2023 MacBookPro with an Apple M2 Pro processor and 16 GB of RAM, with no applications running other than RStudio.

Dataset	Size (features x samples)	Time (seconds ± s.d.)	
		Single core	Multi-core
Yeast transcriptome	5,891 x 86	54 (1.1)	52 (1.6)
Vaginal metatranscriptome	1,658 x 42	10 (0.2)	10 (1.2)
PD1 immunotherapy	20,449 x 109	402 (12.0)	460 (43.5)



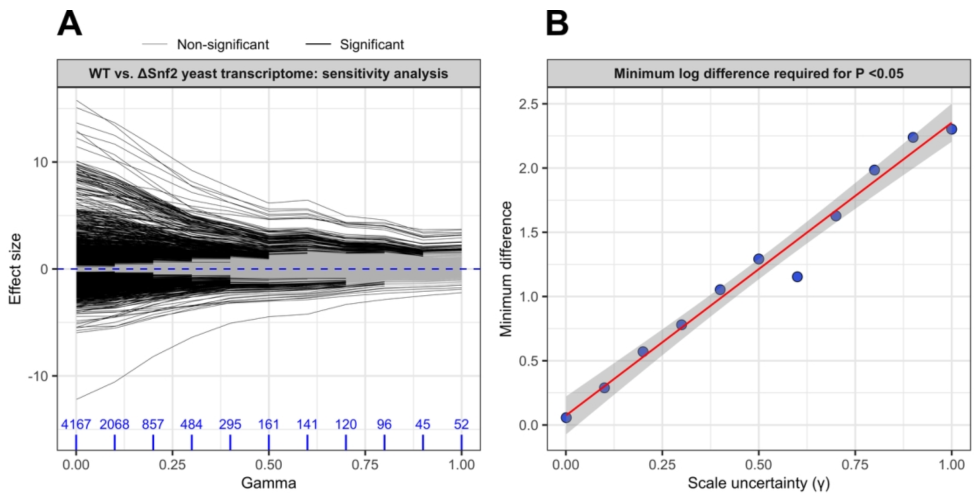
Using a simple scale model eliminates the need for a dual P-value / fold-change threshold: Count data from the yeast transcriptome dataset underwent log ratio transformation using the ALDEx2 package with (A) and without (B) modelling scale uncertainty (gamma = 0.5). Effect sizes and posterior predictive P values were calculated for all features, comparing wild-type and Snf2 knockout cells.

140x114mm (300 x 300 DPI)



Implementing a full, informed scale model corrects data asymmetry in problematic datasets: Count data from the vaginal metatranscriptome dataset was aggregated by KEGG orthology term prior to log ratio transformation using three different approaches: (A) a scale-naïve analysis; (B) a simple scale model using gamma = 0.5; (C) a full scale model using gamma = 0.5 and a difference between conditions of 14 %. Data represented as effect plots. Black points represent non-significant functions, red points represent differentially expressed functions, yellow points represent housekeeping functions. Grey lines indicate equivalent difference within groups and difference between groups. Blue line indicates no difference between groups.

140x56mm (300 x 300 DPI)



Sensitivity analysis informs the selection of an appropriate gamma value: Count data from the yeast transcriptome dataset was used for a sensitivity analysis in ALDEx2, using a range of gamma values from 0 to 1.0 in increments of 0.1. (A) Effect sizes at each gamma value are plotted for all features and coloured by significance at each gamma value. Blue numbers indicate the number of differentially expressed features at each gamma value. (B) The minimum absolute log difference between wild-type and Snf2 knockout cells required for a statistically significant result is shown for each gamma value. Regression line represents a slope of 2.2.

140x70mm (300 x 300 DPI)